

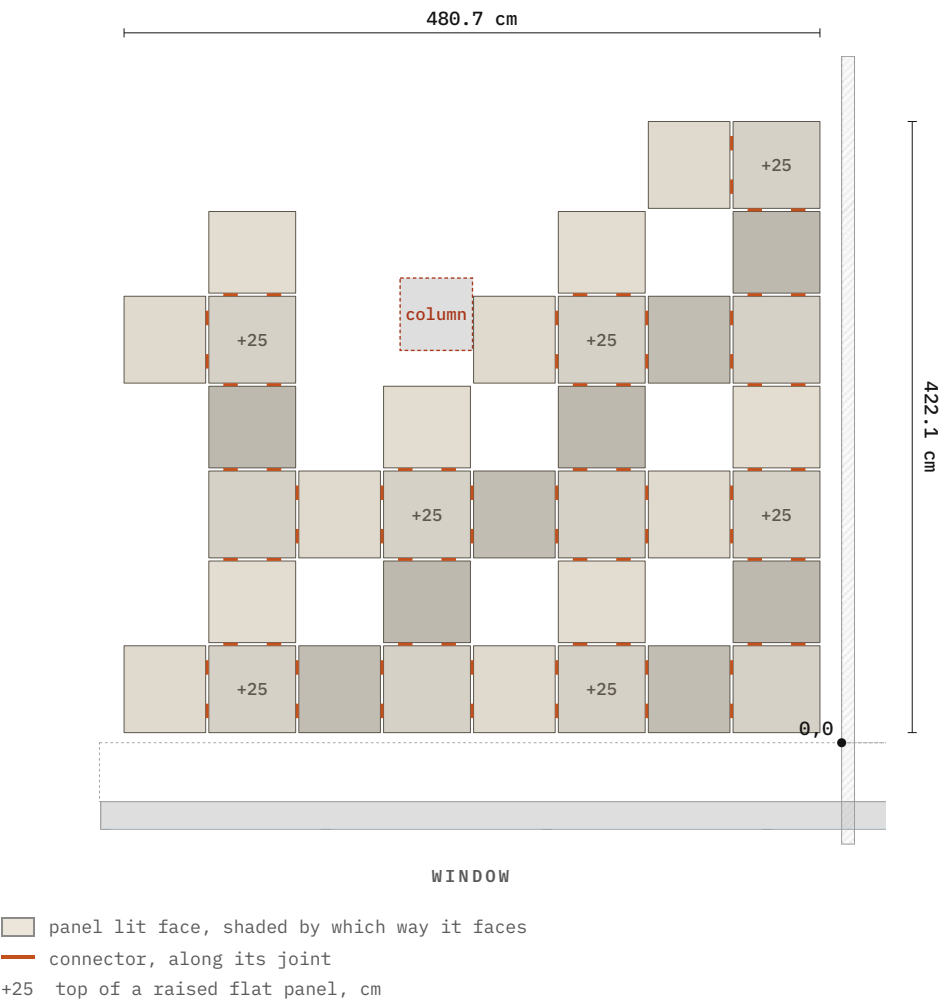
# Structure summary

Design “fold study 1”

|                         |                  |         |           |
|-------------------------|------------------|---------|-----------|
| PATTERN                 | FOOTPRINT        | HEIGHT  | PANEL GAP |
| trapezoid · 5 × 4 cells | 480.7 × 422.1 cm | 25.4 cm | 22 mm     |

|                                    |                                              |                                                  |
|------------------------------------|----------------------------------------------|--------------------------------------------------|
| PANELS                             | INCLINE ANGLE                                | CONNECTORS                                       |
| 34                                 | 20°                                          | 74                                               |
| 22 inclined · 12 flat · 60 × 60 cm | every inclined panel · connector limit 40.6° | 2 per joint · 37 joints · 148 pieces · 222 bolts |

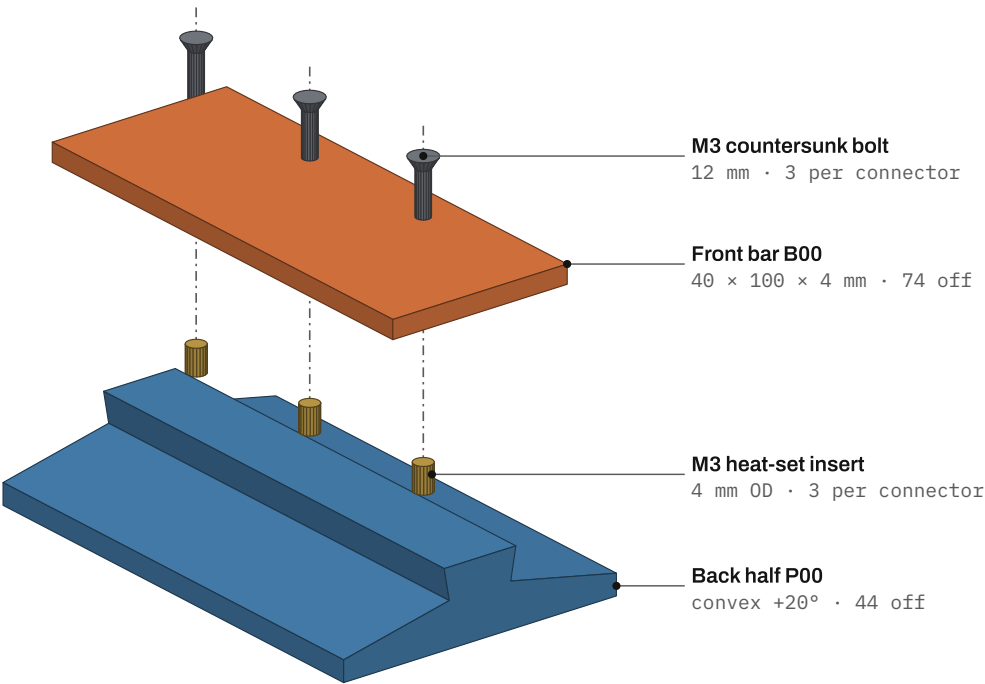
## Plan from above · window at the bottom, wall on the right



| PANELS                                              |                     |     |
|-----------------------------------------------------|---------------------|-----|
| TYPE                                                | LIES                | QTY |
| Ground                                              | flat, on the floor  | 5   |
| Raised                                              | flat, top at +25 cm | 7   |
| Rise                                                | inclined 20°        | 9   |
| Fall                                                | inclined 20°        | 13  |
| Total                                               |                     | 34  |
| CONNECTOR KIT                                       |                     |     |
| PIECE                                               |                     | QTY |
| Back half P00 · convex +20°<br>62.6 × 100 × 16.2 mm |                     | 44  |
| Back half P01 · concave -20°<br>72 × 100 × 11.1 mm  |                     | 30  |
| Front bar B00<br>40 × 100 × 4 mm                    |                     | 74  |
| M3 countersunk bolt<br>12 mm long                   |                     | 222 |
| M3 heat-set insert<br>4 mm OD                       |                     | 222 |
| Printed pieces                                      |                     | 148 |

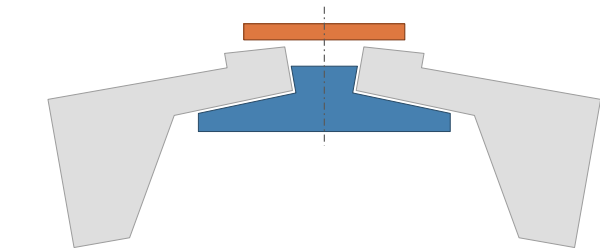
# Connector two printed pieces per joint station

The back half reaches under both panel flanges. The front bar lies across both bezels. Three M3 bolts pull the two together through heat-set inserts, clamping the panel rims between them.

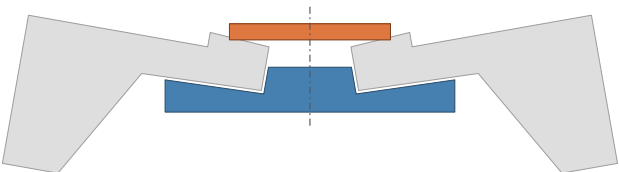


Exploded along the bolt axis. Shown: back half P00 (convex +20°) with front bar B00. Bolt positions are indicative.

## ASSEMBLED, LOOKING ALONG THE JOINT



**P00 · convex +20°**  
10 mm 44 off · built for 20 mm · joints 22 mm



**P01 · concave -20°**  
30 off · built for 20 mm · joints 22 mm

### FRONT BAR FOULS ON CONCAVE JOINTS

A flat bar clears the bezels up to 12.4° of concave fold. This design folds 20°, so the 30 concave connectors (P01) need a different bar section. The overlap shows in the P01 section above.

### PIECES ROUNDED TO THE CONNECTOR BIN

Every joint here is 22 mm, but the kit rounds gaps to 5 mm steps. The back halves are built for 20 mm and the bar is 40 mm wide instead of 42. Set the editor's **span bin** to 0.20 cm or finer to build them at exactly 22 mm.

### HOLES NOT MODELLED

The printable parts are solid. Bolt clearance holes and insert bores still need to be cut, and the three bolt positions along the part are not fixed yet.

### PRINTABLE PIECES

**P00\_back-half\_convex+20deg.stl**  
back half · 62.6 × 100 × 16.2 mm · print 44

**P01\_back-half\_concave-20deg.stl**  
back half · 72 × 100 × 11.1 mm · print 30

**B00\_front-bar\_40mm.stl**  
front bar · 40 × 100 × 4 mm · print 74

Millimetres, laid flat for printing. In grid-designer/sheet/pieces/, or from the editor's Connector pieces button.